Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/27/2008
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Project Title:	Handheld Body-Fluid Analysis System for Astronaut Health Monitoring		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ho, Chih-Ming (University of California, Los Angeles) Kasdan, Harvey (IRIS International)		
Grant/Contract No.:	NCC 9-58-TD00406		
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Task Description:	(1) The specific aim of this project is to focus on the development and demonstration of an automated handheld blood count instrument, which is easy for astronaut to operate. A monolithic chip will be developed to perform measurements of red blood cell (RBC) count, mean cell volume (MCV), hematocrit, white blood cell (WBC) count and at least 2-part differential (lymphocytes verse granulocytes). The system should analyze a minimum of "1,000 RBC" and "200 WBC" which corresponding to processing sample volume ~50-100 nL of whole blood. The sample volume should be precisely measured ± 5%. (2) During the previous funding year, we made progress in electrical impedance sensing. We solved a serious problem for micro electrical impedance sensors, which is the large double layer impedance of electrode-electrolyte interface. The small surface area increases the double layer impedance and lowers the sensitivity. By either electroplating Platinum black on electrode surface or introducing an inductor parallel to the system to induce resonance, we achieve high signal-to-noise ratio when particles including blood cells passing the electrodes. At low AC frequency, the height of the signal was confirmed to be determined by the volume of the particles. The histograms of the signal height for whole blood and leukocyte rich plasma matched very well to previous published volume distributions of erythrocytes and leukocytes. The inductor induced resonance sensing approach can easily enable multi-frequency sensing by introducing multiple inductors with different inductance values to multiple pairs of electrodes along the same microfluidic channel. We successfully demonstrated fluorescent sensing and counting for WBC count and 2-part differential with a portable prototype instrument. We established this method in previous funding years. In brief, cell nucleus fluorescent dye acridine orange stains DNA to be green and RNA to be red. Also this dye is pH sensitive. As a result, non-lymphocytes have a higher red fluorescence level than lymphocytes due to their more active gene expression. So the red fluorescence channel is used for 2-part WBC differential, while the green fluorescent channel is used for WBC count. The unique feature of this technique is that whole blood can be used without dilution, which greatly simplify system design, device operation and buffer storage. The concept was proved last year by constructing an optical detection system on optical bench. Throughput of 1000 WBCs/sec was achieved, which means a WBC count can be finished in one second. In this short period of time, blood cell sedimentation not a problem. During the current year, we successfully made a portable prototype instrument based on this approach. The prototype instrument is made from off-shelf components with customer mechanical design. The system is composed of a blue LED for illumination, a fluorescent microscope filter block as main optical component, and two photon multiplier tubes for fluorescent signal detection. Data from the PMT are amplified by a pre-amp and collected with a data acquisition board inside the instrument. The microfluidic chip is inserted into the sample chamber. A mini peristaltic pump is used to draw the sample through the device at up to 6ÌL/min. Typical time traces before and after the signal passing through a digital low pass filter were recorded. Two clusters corresponding to lymphocytes and granulocytes can be easily identified on the scatter plot. Comparison of lymphocyte percentages from ten clinical samples processed with the prototype and an automated hemacytometer resulted in good correlation. During the past four years, we made significant advance on size-based microfluidic particle separation techniques with application on blood cell separation. Two main types of devices (pillar-shaped and channel shaped) were demonstrated. RBC, WBC, and platelets can be separated into cell populations with over 96% purity. With the help of antibody conjugated beads, we also successfully achieved separation of subtypes of lymphocytes from the rest of the blood cells. (3) The findings above are key components to meet the specific aims of the proposal. Progress in electrical impedance sensing enables us to sense and count RBCs and WBCs with high signal-to-noise ratio, while separation of RBCs and WBCs makes the electrical impedance sensing more sensitive. We also showed under low AC frequency excitation, the signal height of the electrical impedance sensing was determined by the volume of the particle. So with proper calibration, the MCV could be measured directly by measuring individual cell volume and averaging them. Hematocrit can be determined indirectly by adding volumes of individual cells together. Optical fluorescent sensing demonstrated 2-part WBC differential from undiluted blood. Since fluorescent dye AO and anticoagulant EDTA can be coated on channel surface, so sample preparation will be minimized, which will be very convenient for space use. The system integration effort resulted in portable prototype equipment for WBC count and differential. This is the start point for product development of the commercial hand held hemocytometer. (4) This is the last year of the funding period for the current proposal. With the successful completion, we are have recently started working on the renew proposal. The new specific aims include (1) 5-part WBC differential, (2) analysis of WBC subtypes (e.g., CD4+ T helper and natural killer cells), and (3) serum/plasma protein biomarker analysis (e.g., for infection, radiation and bone loss monitoring). Our proposed system has a much smaller size, lighter weight, lower voltage supply, less power, smaller sample volume. The functions provided by our proposed instrument are not available from any single commercially available instrument. More importantly, the proposed functions meet NASA's special needs for space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The devices under development can be used for earth-based applications. The proposed device use cartridge and hand held system. The cartridge will be cheap and disposable. The results will be available almost immediately to the patients without going through central lab facilities. The device can be used in emergency room, on ambulance as well as at home. As the senior population continuous to grow, this kind of device will find more and more appealing in point-of-care applications.
Task Progress:	1. On-chip aspiration: We decided to use miniaturized mini peristaltic pumps for fluidic manipulation. Initial testing confirmed that the operation of the pump meets our requirements. 2. On-chip dilution: For fluorescent detection, we demonstrated that dilution is not needed. For separation and then electrical impedance counting, we demonstrate that separation itself is an effective dilution. 3. RBC MCV: After we solved the large double layer impedance problem inherent to micro impedance sensing, high signal-to-noise ratio signals enabled us to perform statistical analysis on the signal height. We confirmed that at low AC excitation frequency (under 100kHz), the signal height is determined by the volume of the particle. Thus, individual volume of RBC can be determined and MCV can be calculated by averaging. 4. WBC differential: We demonstrated WBC differential by AO staining of WBCs and fluorescent detection at red channel. We are investigating the possibilities of 3-part and 5-part WBC differential by AO staining, which are beyond the original proposed.

	5. Hematocrit: Hematocrit is supposed to be calculated by adding measured individual cell volume. 6. Instrument integration: We made a portable prototype instrument integrated all the parts together for WBC count and differential. The prototype instrument uses disposable microfluidic chip as cartridge. Clinical sample tests of the prototype instrument correlates well with the results from conventional bulk blood analyzer.
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